WORKED SOLUTION

STEP III 1996, Question 5

Pure • De Moivre and Trigonometric Identities

STEP III 1996, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q5

The key idea

Expanding $(\cos\theta + \mathrm{i}\sin\theta)^7$ by the binomial theorem and comparing with de Moivre's theorem gives $\cos 7\theta$ and $\sin 7\theta$ as polynomials; dividing through by $\cos^7\theta$ turns everything into powers of $t = \tan\theta$. Once $\tan 7\theta$ is a ratio of polynomials in t, the angles where $\tan 7\theta$ is zero make the numerator vanish, and the angles where $\tan 7\theta$ is undefined make the denominator vanish. Both polynomials are even apart from a factor t, so they become cubics in t^2, and Vieta's formulae do the rest.

Solution

The formula for $\tan 7\theta$

By de Moivre's theorem and the binomial theorem, with $c = \cos\theta$ and $s = \sin\theta$,

$$\cos 7\theta + \mathrm{i}\sin 7\theta = (c + \mathrm{i}s)^7 = \sum_{k=0}^{7} \binom{7}{k} \mathrm{i}^k c^{7-k} s^k.$$

Suppose $\cos\theta \neq 0$, so that $t = \tan\theta$ is defined. Taking out c^7 gives $(c+\mathrm{i}s)^7 = c^7(1+\mathrm{i}t)^7$. Using $\mathrm{i}^2 = -1$, $\mathrm{i}^3 = -\mathrm{i}$, $\mathrm{i}^4 = 1$ and the binomial coefficients $1, 7, 21, 35, 35, 21, 7, 1$,

$$(1+\mathrm{i}t)^7 = (1 - 21t^2 + 35t^4 - 7t^6) + \mathrm{i}\,(7t - 35t^3 + 21t^5 - t^7).$$

Equating real and imaginary parts (everything except i is real),

$$\cos 7\theta = \cos^7\theta\,(1 - 21t^2 + 35t^4 - 7t^6),$$
$$\sin 7\theta = \cos^7\theta\,(7t - 35t^3 + 21t^5 - t^7).$$

Dividing, the factor $\cos^7\theta$ cancels, and multiplying the top and bottom by -1,

$$\tan 7\theta = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6} = \frac{t(t^6 - 21t^4 + 35t^2 - 7)}{7t^6 - 35t^4 + 21t^2 - 1},$$

as required.

For later, note what the two displayed lines say. Write $N(t) = t(t^6 - 21t^4 + 35t^2 - 7)$ and $D(t) = 7t^6 - 35t^4 + 21t^2 - 1$. Then $\sin 7\theta = -\cos^7\theta\, N(t)$ and $\cos 7\theta = -\cos^7\theta\, D(t)$, so when $\cos\theta \neq 0$,

$$\sin 7\theta = 0 \iff N(t) = 0, \qquad \cos 7\theta = 0 \iff D(t) = 0.$$

Part (i)

Take $\theta = \frac{k\pi}{7}$ with $k = 1, 2, 3$. Then $\sin 7\theta = \sin k\pi = 0$ and $\cos\theta \neq 0$, so $N(\tan\theta) = 0$. Also $\tan\theta \neq 0$, since $0 < \theta < \frac{\pi}{2}$, so we may cancel the factor t:

$$t^6 - 21t^4 + 35t^2 - 7 = 0.$$

So $x = t^2$ satisfies

$$x^3 - 21x^2 + 35x - 7 = 0$$

for $x = \tan^2\frac{\pi}{7},\ \tan^2\frac{2\pi}{7},\ \tan^2\frac{3\pi}{7}$.

Ask yourself How do we know these three numbers are *all* the roots, not one root counted several times?

The angles $\frac{\pi}{7} < \frac{2\pi}{7} < \frac{3\pi}{7}$ all lie in $(0, \frac{\pi}{2})$, where tan is positive and strictly increasing. So the three values of $\tan^2$ are different. A cubic has at most three roots, so these are exactly its roots, and this is the cubic we want.

By Vieta's formulae the product of the roots is 7:

$$\tan^2\tfrac{\pi}{7}\,\tan^2\tfrac{2\pi}{7}\,\tan^2\tfrac{3\pi}{7} = 7.$$

Each tangent is positive, because each angle lies in $(0, \frac{\pi}{2})$, so we take the positive square root:

$$\tan\tfrac{\pi}{7}\,\tan\tfrac{2\pi}{7}\,\tan\tfrac{3\pi}{7} = \sqrt{7}.$$

Part (ii)

Now take $\theta = \frac{\pi}{14}, \frac{3\pi}{14}, \frac{5\pi}{14}$. Then $7\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$, so $\cos 7\theta = 0$, while $\cos\theta \neq 0$. By the remark above, $D(\tan\theta) = 0$, so $y = \tan^2\theta$ satisfies

$$7y^3 - 35y^2 + 21y - 1 = 0.$$

As in part (i), the three angles lie in $(0, \frac{\pi}{2})$ and are different, so the three values of $\tan^2\theta$ are different and are exactly the roots of this cubic. By Vieta's formulae their sum is $\frac{35}{7}$:

$$\tan^2\tfrac{\pi}{14} + \tan^2\tfrac{3\pi}{14} + \tan^2\tfrac{5\pi}{14} = 5.$$

There's a quick alternative using part (i). Since $\frac{\pi}{14} = \frac{\pi}{2} - \frac{3\pi}{7}$, and similarly $\frac{3\pi}{14} = \frac{\pi}{2} - \frac{2\pi}{7}$ and $\frac{5\pi}{14} = \frac{\pi}{2} - \frac{\pi}{7}$, the required sum is $\sum 1/x_i$ over the roots x_i of the cubic in part (i). That is

$$\frac{x_1x_2 + x_2x_3 + x_3x_1}{x_1x_2x_3} = \frac{35}{7} = 5.$$

Equivalently, substituting $x = 1/y$ in the part (i) cubic and clearing denominators gives the part (ii) cubic.

Checks

- Put $t = 1$, that is $\theta = \frac{\pi}{4}$. The formula gives $\frac{1-21+35-7}{7-35+21-1} = \frac{8}{-8} = -1$, and indeed $\tan\frac{7\pi}{4} = -1$.
- For small θ, $\tan 7\theta \approx 7t$, and the formula gives $\frac{-7t}{-1} = 7t$ to first order.
- Numerically, $\tan\frac{\pi}{7}\tan\frac{2\pi}{7}\tan\frac{3\pi}{7} \approx 0.4816 \times 1.2540 \times 4.3813 \approx 2.6458 \approx \sqrt{7}$, and $\tan^2\frac{\pi}{14} + \tan^2\frac{3\pi}{14} + \tan^2\frac{5\pi}{14} \approx 0.0521 + 0.6360 + 4.3119 = 5.0000$.

Common slips

- Forgetting to justify that the three values are distinct, so that they really are *all* the roots of the cubic.
- Answering $\pm\sqrt{7}$ in part (i): the sign is fixed because all three tangents are positive.
- Sign errors from the powers of i: $\mathrm{i}^6 = -1$ and $\mathrm{i}^7 = -\mathrm{i}$.

Beyond STEP

The numbers $\tan^2\frac{k\pi}{7}$ are conjugate algebraic numbers: they are the roots of one irreducible cubic with integer coefficients, so any symmetric expression in them is rational. This is the starting point of Galois theory, which studies the symmetries between the roots of a polynomial. The same trick also gives identities such as $\sum_{k=1}^{n}\cot^2\frac{k\pi}{2n+1} = \frac{n(2n-1)}{3}$, used in one elementary proof that $\sum 1/k^2 = \pi^2/6$. For a geometric view of de Moivre's theorem and roots of unity, see Needham, *Visual Complex Analysis*; for more problems of this kind, see Siklos, *Advanced Problems in Mathematics*.

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